

Sealed Experiment Record

Sealed 2026-10-10 · Signed by the K-Veritas server
Session 03e494aa · Data hash 224d...bc96

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-10 19:08 and 19:11 UTC. Reported results: final_test_accuracy 99.35, final_test_loss 0.0281. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	03e494aa-7420-43d8-b526-8f7b4f14501c
Initialized	2026-10-10T19:08:47Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	2de150af2e1ebb74cbbb0eedb98635cfb4bd5672aa54dbe8767decd732fadb2e
Disclosure	open
Source code, run 1	4ef46351617452a4c5090860065255b56ec8a9aaad1e7061475dab594f009d01

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	2 minutes 38 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
test_loss	0.0281	14 (14 readings)	explicit
test_accuracy	99.35	14 (14 readings)	explicit

Inline claim	Value	Phase	Output line
final_test_accuracy	99.35		73
final_test_loss	0.0281		74

Seed committed: seed:1, output line 1, 2026-10-10T19:08:47Z.

4 Execution evidence

Coherence

Coherent (0.38): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	1574	106.4 s	3.11 GB	16%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 22 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	19:08:47	a98c50e637d7b63e	22
2	run_end	19:11:25	a98c50e637d7b63e	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Activity: 26 files read, 0 files written, 32 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-10T19:11:26.569451+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	224d6b79199f01e618e7a65ba5597a782d5839827c182ceb2dcc35880e0fbc96
Code bundle hash	1a8c1c9f0a0f386f33986fbc6bfb2c2d6d8fdc2fbd7b3dfb565a81a36aed77c

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
test_loss	0.0391	1	45	explicit
test_accuracy	98.58	1	46	explicit
test_loss	0.0285	2	47	explicit
test_accuracy	99.07	2	48	explicit
test_loss	0.0254	3	49	explicit
test_accuracy	99.26	3	50	explicit
test_loss	0.0236	4	51	explicit
test_accuracy	99.28	4	52	explicit
test_loss	0.0235	5	53	explicit
test_accuracy	99.27	5	54	explicit
test_loss	0.0227	6	55	explicit
test_accuracy	99.29	6	56	explicit
test_loss	0.0231	7	57	explicit
test_accuracy	99.36	7	58	explicit
test_loss	0.0229	8	59	explicit
test_accuracy	99.35	8	60	explicit
test_loss	0.0246	9	61	explicit
test_accuracy	99.35	9	62	explicit
test_loss	0.0264	10	63	explicit
test_accuracy	99.32	10	64	explicit
test_loss	0.0269	11	65	explicit
test_accuracy	99.33	11	66	explicit
test_loss	0.0274	12	67	explicit
test_accuracy	99.35	12	68	explicit
test_loss	0.0278	13	69	explicit
test_accuracy	99.35	13	70	explicit
test_loss	0.0281	14	71	explicit
test_accuracy	99.35	14	72	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	2	19:08:47	0.5 s	0.58 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	19:08:47	a98c50e637d7b63e	22
1	phase train	19:08:47	a98c50e637d7b63e	0
2	run_end	19:11:25	a98c50e637d7b63e	0

Event	Path or command	Hash
read	kv_run.sh	
read	examples/mnist/main.py	

[illegible]

Event	Path or command	Hash
process	python -u main.py	
process	python -u main.py	
process	python -u main.py	
process	python -u main.py	
process	python -u main.py	
process	pid 2721632	

C Run history

Invocation 1, OK, 2 minutes 38 seconds, 74 output lines, started 2026-10-10T19:08:47

metric_hash fa93e219501db9449e55ae3252098c526b0267d0e1c43ba15e7e03f8ffef4e69
run_digest e78be7c5755d2713a105a6f76cfdde8e014a86c917445acecd31aeb2f0035f40
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 224d6b79199f01e618e7a65ba5597a782d5839827c182ceb2dcc35880e0fbc96
signed_message_hash 937bea75974a2bbbd5ebc953a0f292e383aba88a2fdbb4f442dbd557936313a4
nonce 91c89d3eac6e8e1221711de8013fb062
signed_at 2026-10-10T19:11:26.569451+00:00
sealed_at 2026-10-10T19:11:26.63542714Z
checkout_bundle 1a8c1c9f0a0f386f33986fbc6bfbe2c2d6d8fdc2fbd7b3dfb565a81a36aed77c

Signature (base64)

SCicYcDMivQZbBH7T78aCOX4g3sYXMBEOV+HWrPU3ht5BiVv0lniUwo9im5AUSCiPNJ/dIUV79dvAOK6TQCeHu68
0Zy7F08RkoVjl5oeFXAvqLBRUKbYQCuLwFgHoTdC1xR5KZcCBkzActCvrl3QvGmi/ay04UPqdcncnDWc5Jp923xn
El+0uELfZXda5QjIOjUxTuZn9UVHuK19H8CpYoiTk0HyrUnaCdaXoY04Hlv2xibxQAew4dpAxlN7rH0W0FMbr3
gHRPDg8AHCQvPXshW7EziTPTfv5M+Q03eo5G5vZz0K0Q5voX8owAk0NfttTiK0HXGtc3sg3l3NjQ00f6dn5qULsh
l0LLDMMLEcXSk11N9p/gfr5eagsxtChq7JjsSRJX7EWhp9CU8muDs+dwQZp3QcuWM6ot9Z62hmgkviBAzqCH3pF1
rO/Xv5dE5AxL6DIMivUYhdMz0YqhyUABpSe5ZUwkP0CXftwwX1VvyFY/BtyGi/QwnweHyNSKBuylQI1imTr8Au3
PbT9eL93BEV5eHXsSQAd013WMQos6s9UPa2b799w3xZA1ikg5TjLyeEVqY1E4SjBxYGF68G9GuWYLP+HyaVIDcqs
p1lP2EHo0FiH5GHiAZD+ra7wFU7j/l0UteFQ1ba0cvMjXDS+00SeB0qeZ5C2uufX0Us=

Public key

-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCGKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekm1PlcjayJxD5fcPrOPPCqp/LoUFz5CUZtRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNPpxYa5X
7qU95EfEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KIcEadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNfCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdhQ+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9IL5xEaJn4bku4eKQeS7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7LE3mbQRBpSLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PwjNcNku5cdw3gHLcCAwEAAQ==
-----END PUBLIC KEY-----

Run 1 outputs

stdout 2ef6d2344315fa34cb054bea7cf08c2634a23ead43650c11841356df710020bb
stderr e3b0c44298fc1c149afbfc4c8996fb92427ae41e4649b934ca495991b7852b855
metrics fa93e219501db9449e55ae3252098c526b0267d0e1c43ba15e7e03f8ffef4e69
digest e78be7c5755d2713a105a6f76cfdde8e014a86c917445acecd31aeb2f0035f40
kv_run.sh (unchanged)
pre feb85e6349e7996c2ea927457942c5318ebc71f8aeb414ace064f83f74534dba

post feb85e6349e7996c2ea927457942c5318ebc71f8aeb414ace064f83f74534dba

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```